Related Rates and Optimization: Diagnosing the Setup Answer Key

AP Calculus AB / BC — Applications of Derivatives

Quick Answers

1. $2\pi r$
2. $2\pi r^2$
3. 0.318
4. -1.5
5. -0.955
6. $100 - x$, 100
7. $12x^2 - 192x + 576$, 4, 12
8. —

Curriculum

Level: AP Calculus AB / BC — Applications of Derivatives
CHA-3.D – *problems 1, 2, 3, 4, 5*
FUN-4.B–FUN-4.C – *problems 6, 7, 8*
Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

3. *If they got 1.59: used the circumference-style factor 4*pi*r instead of the surface factor 4*pi*r^2*
4. *If they got -2.67: inverted the ratio, using the wall height over the ground distance instead of the other way round*
5. *If they got -1.27: held the radius fixed at 3 instead of substituting $r = h/2$ before differentiating*

1. Differentiate $A = \pi r^2$ with respect to r.

Solution: π is a constant, so the power rule applies to r^2 alone: $\dfrac{dA}{dr} = \pi \cdot 2r$. $\boxed{2\pi r}$

This is the factor the chain rule attaches to $\dfrac{dr}{dt}$: $\dfrac{dA}{dt} = 2\pi r \dfrac{dr}{dt}$. Note that it still contains r — that is exactly why a numeric radius cannot be substituted before this step.

2. Cone with $h = 2r$: eliminate, then differentiate.

Solution: Substitute the geometric constraint into the volume relation first, so that only one variable remains:

$$V = \tfrac{1}{3}\pi r^2 h = \tfrac{1}{3}\pi r^2 (2r) = \tfrac{2}{3}\pi r^3$$

Now differentiate: $\dfrac{dV}{dr} = \tfrac{2}{3}\pi \cdot 3r^2$. $\boxed{2\pi r^2}$

Why the order matters: differentiating $\tfrac{1}{3}\pi r^2 h$ directly would need a product rule and a second unknown rate $\dfrac{dh}{dt}$. Eliminating first is what makes the problem solvable with one given rate.

3. Balloon: $\dfrac{dV}{dt} = 100$, find $\dfrac{dr}{dt}$ at $r = 5$.

Solution: Relation and differentiation come before any number:

$$\begin{aligned} V &= \tfrac{4}{3}\pi r^3 \\ \frac{dV}{dt} &= 4\pi r^2 \frac{dr}{dt} \\ \frac{dr}{dt} &= \frac{dV/dt}{4\pi r^2} \end{aligned}$$

Only now substitute the instantaneous values $\dfrac{dV}{dt} = 100$ and $r = 5$: $\dfrac{100}{4\pi(5)^2} = \dfrac{100}{100\pi} = \dfrac{1}{\pi}$.

$\boxed{0.318}$

Watch for: 1.59 comes from dividing by $4\pi r$ rather than $4\pi r^2$ — the derivative factor was taken from the wrong power. A quick sanity check: the divisor must have the units of an area, and $4\pi r$ does not.

4. Ladder: 10 ft, base at 6 ft moving out at 2 ft/s.

Solution: Let x be the base distance and y the wall height. The ladder's length is the constant here, and x and y both vary:

$$\begin{aligned} x^2 + y^2 &= 10^2 \\ 2x\frac{dx}{dt} + 2y\frac{dy}{dt} &= 0 \\ \frac{dy}{dt} &= -\frac{x}{y} \cdot \frac{dx}{dt} \end{aligned}$$

Solve for the wanted rate symbolically first, then substitute. At the instant in question $x = 6$, so $y = \sqrt{10^2 - 6^2} = 8$, and $\dfrac{dy}{dt} = -\dfrac{6}{8}(2)$. $\boxed{-1.500}$

The negative sign is part of the answer: the top slides down. *Watch for:* -2.67 is $-\dfrac{y}{x}\dfrac{dx}{dt}$, the inverted ratio. Solving the differentiated equation symbolically before substituting is what prevents it.

5. Find and fix: the conical tank with a "fixed" radius.

Solution: Writing $\frac{dV}{dt} = \frac{1}{3}\pi r^2 \frac{dh}{dt}$ with r held at 3 assumes the water's surface keeps the radius of the tank's rim as the level falls. It does not: the surface is a cross-section of the cone, so it shrinks with the depth. Similar triangles give the constraint $\frac{r}{h} = \frac{3}{6}$, i.e. $r = \frac{h}{2}$. Substitute it, then differentiate:

$$V = \tfrac{1}{3}\pi\left(\tfrac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$$
$$\frac{dV}{dt} = \frac{\pi h^2}{4} \cdot \frac{dh}{dt}$$
$$\frac{dh}{dt} = \frac{4}{\pi h^2} \cdot \frac{dV}{dt}$$

With $\frac{dV}{dt} = -12$ (draining) and $h = 4$: $\frac{dh}{dt} = \frac{4(-12)}{\pi(4)^2} = -\frac{3}{\pi}$. $\boxed{-0.955}$

Watch for: -1.27 is what the student's fixed-radius setup produces. It is too fast by a factor of $\frac{4}{3}$, and the sign of the error is predictable: pretending the surface stays wide makes the level appear to drop faster than it does.

6. Find and fix: the pen against the barn.

Solution: The constraint $2x + 2y = 200$ is the perimeter of a closed rectangle, but the barn supplies one side, so only three sides are fenced. Let x be the side parallel to the barn and y each of the two perpendicular sides. The correct constraint is $x + 2y = 200$, giving $y = \frac{200 - x}{2}$. Now write the area with one variable and differentiate:

$$A(x) = x \cdot \frac{200 - x}{2} = 100x - \frac{x^2}{2}$$
$$A'(x) = 100 - x$$

Setting $A'(x) = 0$ gives the critical value. Since $A''(x) = -1 < 0$, it is a maximum.

$$\boxed{A'(x) = 100 - x \qquad x = 100}$$

The corresponding y is 50 and the maximum area is 5000 square feet. The student's constraint would have given $x = 50$ and an area of 2500 — exactly half, because their pen spends fencing on a side the barn already provides.

7. Open box from a 24 by 24 sheet.
Solution: Cutting a square of side x from each corner leaves a base of side $24 - 2x$ and a height of x:

$$V(x) = x(24 - 2x)^2 = 4x^3 - 96x^2 + 576x$$
$$V'(x) = 12x^2 - 192x + 576 = 12(x^2 - 16x + 48) = 12(x - 4)(x - 12)$$

$$\boxed{V'(x) = 12x^2 - 192x + 576 \qquad x = 4 \quad x = 12}$$

Rejecting a critical value: the physical domain is $0 < x < 12$, because two cuts of side x must fit across a side of length 24. At $x = 12$ the base has side $24 - 2(12) = 0$, so the box has no volume — it is an endpoint of the domain, not an interior maximum. The maximum is at $x = 4$, where $V = 4(16)^2 = 1024$. Stating the domain *as part of the setup* is what makes this rejection automatic rather than an afterthought.

8. Explain: substituting before differentiating. *(Open response — flagged for manual review; a model answer follows.)*
Model answer: Substituting the instantaneous radius into $V = \frac{4}{3}\pi r^3$ turns a function of a changing quantity into a single number. Differentiating a number gives zero, so the conclusion "the rate is zero" is a faithful consequence of the wrong first step, not an arithmetic slip. The relation must hold at *every* instant, not just the one of interest, because differentiation compares nearby instants; freezing r discards the very variation the derivative measures.
The distinction to keep is between quantities that vary and quantities that are genuinely constant. In the balloon, r and V both vary with time and neither may be replaced by a number before differentiating. In the ladder problem, the ladder's length is a true constant, which is why it may be written as 10 from the start while x and y may not. A useful test: if the quantity has a non-zero rate of change in the problem, it must stay a symbol.
Substitution becomes legal once differentiation is finished and the equation has been solved for the desired rate. At that point the equation relates rates *at a general instant*, and putting in the values for one instant evaluates it there.
What a reviewer should look for: the response must separate varying quantities from genuine constants and must state that substitution is legal only after differentiating.